Louis Zhang

MECHANICAL ENGINEERING STUDENT - POLYTECHNIQUE MONTREAL

Profile

Mechanical Engineering Student at Polytechnique Montréal with a track record of delivering high-impact prototypes — from lunar transport systems to medical assistive robotics. I bridge the gap between mechanical design and intelligent control to solve real-world engineering challenges through iterative prototyping and technical rigor.

Contact

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- LinkedIn
- Github

Core Compentencies

Domain	Tools & Comptencies
Design	CATIA V6, SolidWorks, GD&T (ASME Y14.5), DFM/DFA, Blender
Analysis	FEA (Ansys, SolidWorks Simulation), MATLAB/Simulink, CFD
Robotics	ROS 2, Python, C++, Sensor Fusion (mmWave, IMU, LIDAR)
Hardware	Arduino, ESP32, Pi Pico, CNC Machining, Rapid Prototyping
Software	Python, C++, React.js, MongoDB, Javascript
Project Management	MS Project, BOM, CSA Compliance Documentation

Portfolio Contents

3
Projects

2x
Hackathon wins

Internships

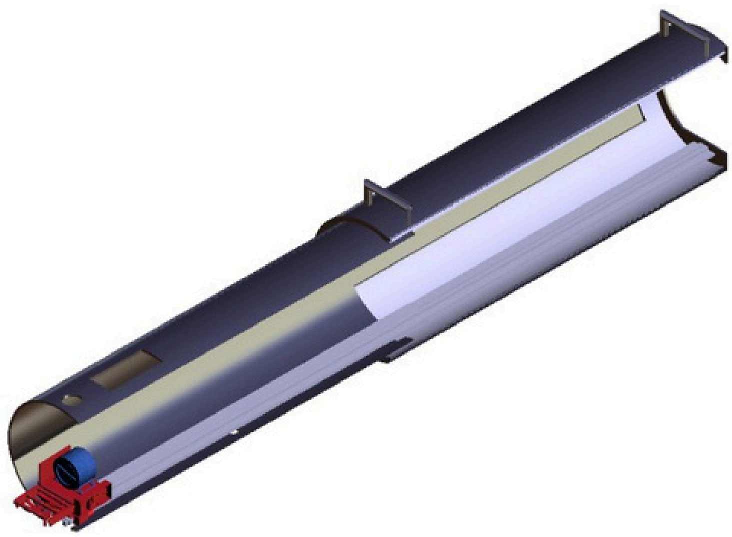
Tesla
Manufacturing Engineering Intern
June 2026 - Aug 2026

Lockheed Martin
Mechanical Engineering Intern
January 2026 - April 2026

City of Montreal
Inspector
May 2025 - August 2025

CSA LEAP PROJECT PORTFOLIO

LUNAR TRANSPORT SYSTEM



Section Cut CAD — Interior Detail, 3.1m System

- Directed end-to-end conceptualization and holistic integration of a **3.1 m lunar produce-transport system** under the Canadian Space Agency LEAP program. Orchestrated a cross-disciplinary engineering team from concept through physical build to meet **stringent CSA safety profiles.**

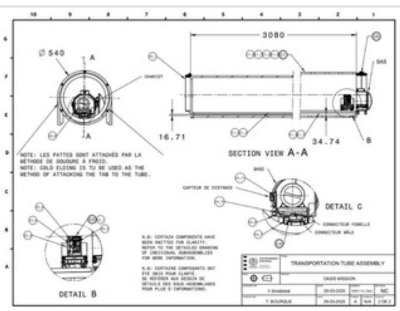
3.1m	-12%	2.5×	50+
Telescopic Envelope	Mass Reduction	Safety Factor	Part Interfaces

PRIMARY TOOLSET

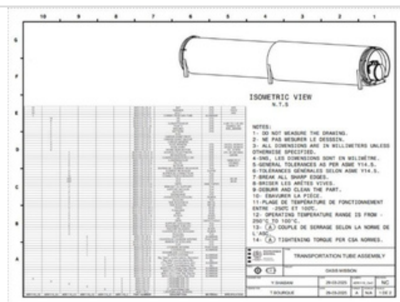
CATIA V6 · SolidWorks · Fusion 360 · GD&T · MS Project · Cold Welding

DESIGN ARCHITECTURE

Constraint Management · Production Specification



TRANSPORTATION TUBE ASSEMBLY — SECTION VIEW A-A



TRANSPORTATION TUBE ASSEMBLY — ISOMETRIC VIEW & BOM

Constraint Management

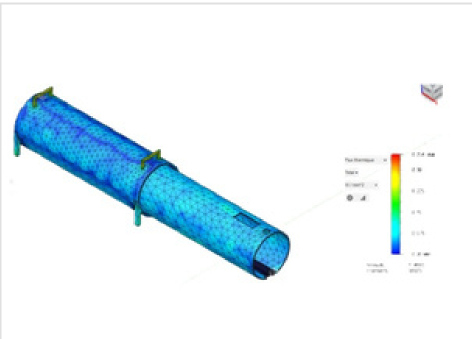
- Managed **50+ unique part interfaces** across the full system envelope — coordinating geometry conflicts between telescopic, structural, andthermal subsystems
- Iteratively optimized part geometry to achieve a **12% system mass reduction** while sustaining a **2.5× structural safety factor** under lunar load conditions
- Used **CATIA V6** parametric assemblies to enforce interface constraints and propagate design changes across dependent geometry
- Coordinated design freeze milestones with a **cross-disciplinary team** using MS Project, tracking interface dependencies across mechanical, thermal, and electrical subsystems

Production Specification

- Applied **GD&T** to all critical mating surfaces — defining flatness, perpendicularity and true-position tolerances for downstream manufacturing
- Defined **cold-welding assembly parameters** for tab-to-tube joints to eliminate thermal fatigue failure points in the vacuum operating environment
- Produced formal **manufacturing drawings** and assembly sequences aligned to CSA build standards
- Conducted tolerance stack-up analysis across all telescopic stages to verify fit under worst-case dimensional variation

FEA · THERMAL · KINEMATIC

PREDICTIVE SIMULATION & ANALYSIS

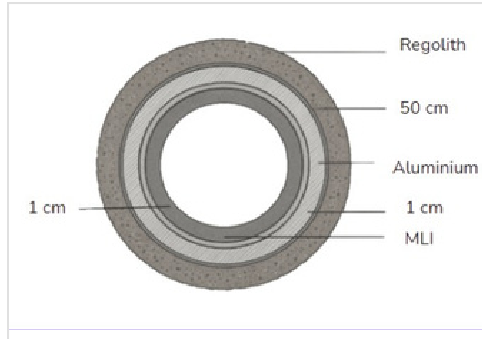


THERMAL FEA
Heat distribution — 14-day lunar night cycle simulation

VALIDATION MATRIX

MAX THERMAL GRADIENT	290°C Delta
STRUCTURAL SF (SAFETY)	2.54x
PREDICTIVE VARIANCE	< 4.2%

VALIDATION MATRIX
Zero-interference validation across full 3.1m deployment



MATERIAL STACK CROSS-SECTION
High-vibration launch profile — drive mechanism modal

SIMULATION ANALYSES

Thermal Interface Control · Kinematic Integrity

Thermal Interface Control

- Verified **heat retention limits** during 14-day lunar night cycle — ensuring produce viability through the full cold-soak period
- Identified and mitigated **thermal bridging** in telescopic aluminum joints through geometry modifications and MLI placement
- Iterative FEA-guided redesign to isolate conduction paths at each joint interface

Kinematic Integrity

- Validated drive mechanism against **high-vibration launch profiles** — confirmed structural integrity under worst-case launch loads
- Verified **zero mechanical interference** across the full 3.1m telescopic deployment sweep through kinematic motion study
- Confirmed clearance margins at every discrete extension state across the deployment range

MATERIAL STACK-UP ENGINEERING

Outer → Inner

Layer 01 — Regolith Shielding

50cm radiation protection layer utilizing **local lunar regolith** for maximum passive thermal mass. Zero additional launch weight penalty — harvested in-situ.
THICKNESS: 50cm · SOURCE: In-situ harvest · FUNCTION: Radiation + thermal mass

Layer 02 — Al 6061 Pressure Vessel

High-strength structural core machined from **Aluminum 6061** — designed for internal atmospheric containment and primary load-bearing under operational pressure.
MATERIAL: Al 6061-T6 · FUNCTION: Structure + pressure containment · SAFETY: 2.5× factor

Layer 03 — Multi-Layer Insulation (MLI)

Reflective barriers optimized to minimize **radiative heat transfer** in high-vacuum operating conditions. Positioned at all critical interfaces to break thermal bridging paths.
TYPE: MLI blanket · FUNCTION: Radiative barrier · ENVIRONMENT: High-vacuum

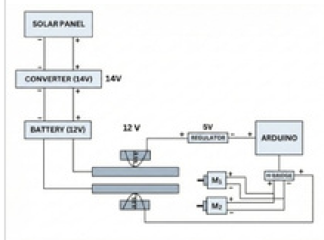
EMBEDDED LOGIC · POWER · INTEGRATION

MECHATRONICS & FABRICATION

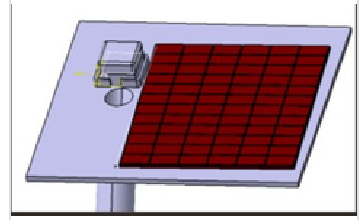
MECHATRONICS & EMBEDDED LOGIC

Power & Charging Interface

Designed the power routing and charging interface to maintain **electrical continuity across all telescopic extension states** — from stowed to full 3.1m deployment. Validated signal integrity throughout the complete motion envelope.



POWER & CHARGING CIRCUIT



SOLAR PANEL ASSEMBLY

DYNAMIC VALIDATION — CONFIRMED

Electrical continuity verified during live telescopic extension cycle. **Autonomous transport cycle** end-to-end — no manual intervention required.

LOGIC STACK INTEGRATION

PRIMARY: ARDUINO (WAGON LOOP)

SECONDARY: PI PICO (CONVEYOR LOGIC)

▶ DYNAMIC VALIDATION — VIDEO 1

Rail Conductivity Test

Validates electrical continuity across telescopic rail extension

WATCH VIDEO →

RAIL CONDUCTIVITY TEST

▶ DYNAMIC VALIDATION — VIDEO 2

Full Assembly Demo

End-to-end autonomous transport cycle — full system operation

WATCH VIDEO →

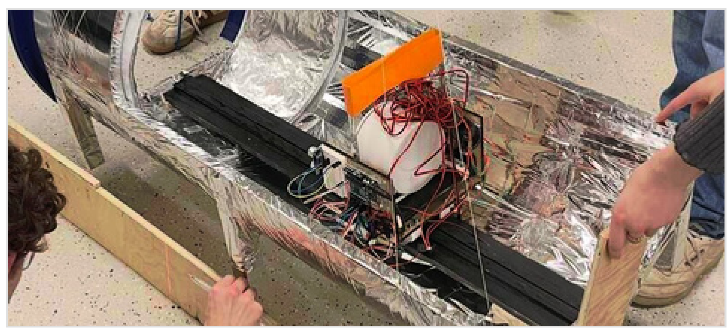
FULL ASSEMBLY DEMO

PHYSICAL FABRICATION & ASSEMBLY

Served as **Lead Physical Integration** — coordinating mechanical, electrical, and thermal assembly sequences across all subsystems to meet CSA build specifications and test readiness milestones.

FABRICATION SPECIALIZATIONS

- AL-6061 Machining
- MLI Fabrication
- PCB Prototyping
- Wiring Harnesses



PHYSICAL INTEGRATION — LAB BUILD

TECHNICAL SUMMARY

At a glance

DISCIPLINE	SCOPE	OUTCOME
Systems Integration	50+ part interfaces, 3.1m envelope	−12% mass, 2.5× safety factor
Thermal Engineering	14-day lunar night cycle FEA	Heat retention verified, bridging mitigated
Mechatronics	PCB, wiring harness, embedded logic	Continuity validated across all states
Fabrication	AL-6061, MLI, PCB, wiring	Lead physical integration — CSA spec

KEY COMPETENCIES DEMONSTRATED

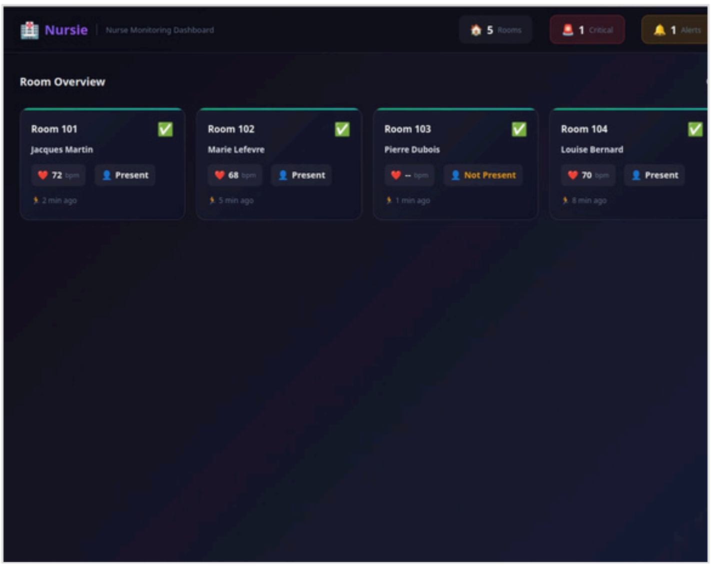
Summary

- ✓ Managed 3.1m telescopic deployment envelope under CSA safety constraints
- ✓ Advanced CAD modelling (CATIA V6) + FEA simulation (thermal & kinematic)
- ✓ Validated electrical continuity across all telescopic motion states
- ✓ Hands-on fabrication lead — machining, PCB, MLI, wiring, full assembly

HACKATHON PROJECT · LONG-TERM CARE · SENSOR FUSION · EDGE AI

🏆 BEST USE OF ELEVENLABS (ConUHacks X)

NURSIE
AI NURSE



Nurse Monitoring Dashboard — Real-time room overview & active alerts

Built an **on-premises AI nurse assistant** for Quebec long-term care facilities (CHSLDs) that detects falls and monitors residents using privacy-preserving hardware — no cameras, no cloud. Falls among seniors (65+) are a leading cause of injury hospitalizations in Canada, rising **47% between 2008–2019**. Nursie addresses the gap between unwitnessed falls and delayed staff response.

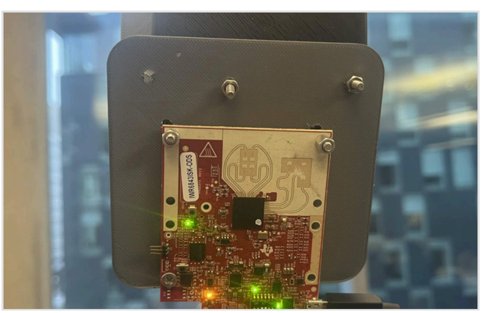
47%	3×	0	LAN
RISE IN FALL HOSPITALIZATIONS	SENSOR REDUNDANCY	CAMERAS USED	LOCAL-ONLY ARCHITECTURE

PRIMARY STACK

`mmWave Radar` `ESP32 / FreeRTOS` `ROS 2` `MongoDB` `Gemini AI` `ElevenLabs` `OpenWRT`

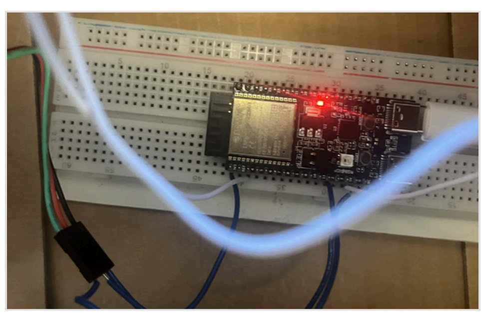
SENSOR HARDWARE

Privacy-Preserving · Redundant · Edge-Processed



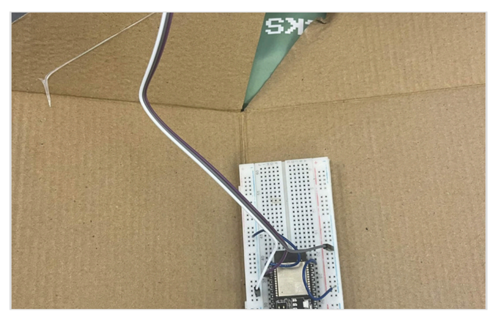
MMWAVE RADAR (IWR6843)

Continuous position & activity sensing via electromagnetic waves. Detects falls and tracks presence with no camera. Signal-processing tuned to reject curtain & object motion noise.



ESP32 DOOR + PIR NODE

Magnetic door sensor + PIR presence detector running FreeRTOS on ESP32. Publishes occupancy state over ROS 2 — flags night-time room exits for anomaly detection.



FLOOR VIBRATION MAT NODE

Detects floor-impact signature on contact. Fused with radar to distinguish a critical fall from a dropped object — reduces false-positive alerts for nursing staff.

DESIGN PRINCIPLES

Architecture Rationale

01 — FAST DETECTION FIRST

High confidence + fast alerting is the system's core mission. Every design choice subordinates itself to detection latency and reliability.

02 — SENSOR REDUNDANCY

mmWave radar, floor vibration mat, and wearable fused together. No single sensor failure breaks detection. Confidence scoring gates alert escalation.

03 — PRIVACY BY ARCHITECTURE

All data stays on a local OpenWRT LAN. No camera. No raw data leaves the facility by default. Compliant with Québec Law 25 privacy framework.

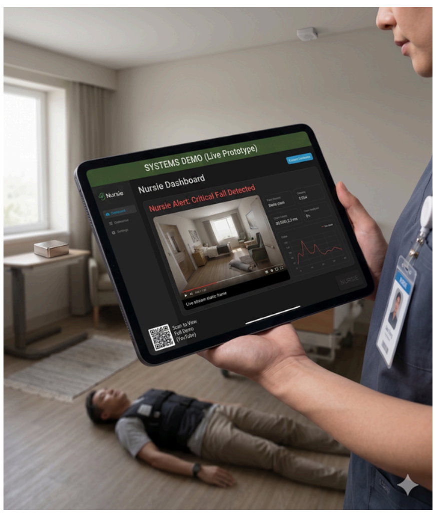
AI PIPELINE · DASHBOARD · OUTCOMES

ALERT PIPELINE & OUTCOMES

AI & BACKEND PIPELINE

Gemini · ElevenLabs · MongoDB · ROS 2

SENSORS	→	EDGE NODE	→	AI TRIAGE	→	PERSISTENCE	→	ESCALATION
mmWave Radar Vibration Mat Door / PIR Smartwatch	→	ESP32 FreeRTOS ROS 2 Messages OpenWRT LAN Signal Fusion	→	Gemini AI Intent / Priority ElevenLabs Voice Webhook Events	→	MongoDB Profile Anomaly Flags Behavior History Triage Logs	→	Phone Call Alert Dashboard Push Critical Priority Nurse Response



Watch Full Demo (Youtube Video)

Project Key Outcomes (Impact & Compliance)

- **Law 25 & Privacy-First:** Satisfies Québec's personal information protection framework (on-premises, zero-cloud dependency).
- **Advanced SNR & Fall Detection:** Fuses mmWave radar + vibration mat for soft vs. critical fall discrimination. Optimized Signal-to-Noise Ratio (SNR).
- **Production-Ready Edge Design:** Scalable, low-latency design on OpenWRT LAN. LAN-local by design.

The Engineering Story: Challenges & Solutions

SIGNAL CLARITY (mmWave SNR Optimization)

Differentiating human heartbeat signatures from moving curtains. Custom signal processing algorithms and threshold tuning.

Impact: Repeatable sensor node performance and manufacturable design.

REAL-TIME RELIABILITY & LAN MESH

Resolving real-time telemetry instability and sensor fusion logic through an optimized ESP32 mesh network.

Impact: Integrated sensor node ready for deployment in high-reliability environments.

Future Roadmap

Apple Watch Integration

Secondary vital-sign validation for high-risk patients.

Predictive Gait Analysis

Using radar data models to estimate fall risk before an incident occurs.

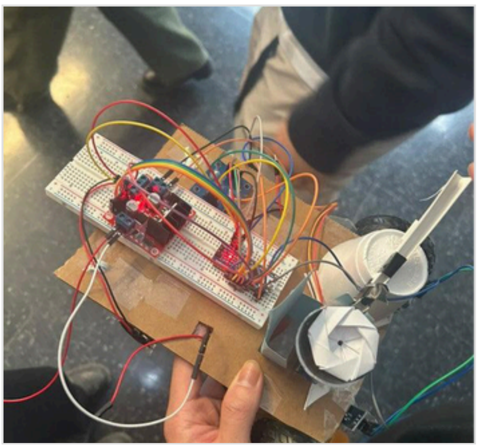
Medical-Grade Hardware

CHSLD rollout plan, with a focus on manufacturing efficiency and scalability.

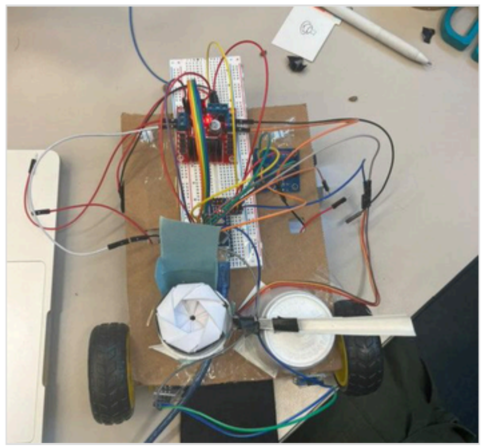
ROBOHACKS · FEB 2025 · ARDUINO · RECYCLED MATERIALS

🏆 WINNER — UNEXPECTED EXPEDITION AWARD · **5th Place Overall**

AUTONOMOUS REFORESTATION ROBOT



Top view — breadboard, Arduino, IR sensors & seed dispenser



Presentation demo — full assembly on competition floor

Built a fully autonomous seed-planting robot in **24 hours** using recycled materials—cardboard chassis, plastic bottles, and a paper-engineered seed dispenser. Coded the entire navigation and dispensing sequence in **C++ on Arduino**, wiring all electronics from scratch.

24h	3	5th
BUILD WINDOW	SENSORS INTEGRATED	PLACE OVERALL

STACK

Arduino C++ · IR Sensors · Color Sensor · DC Motors · Recycled Materials

ENGINEERING & CODE

Arduino · Sensors · Navigation

- **Autonomous navigation** — two infrared sensors fused with a color sensor for precise line-following along a designated black track
- **Motorized seed dispenser** — custom-engineered from paper and recycled materials; servo-actuated for accurate, timed seed placement
- **Full C++ codebase** — sensor polling loop, motor PID steering, and dispenser trigger sequence all written and debugged within the 24-hour window
- **Wire integration** — manually routed all electronics on a breadboard; confirmed signal integrity under live competition conditions

SUSTAINABLE DESIGN

Recycled-First Construction

RECYCLED-FIRST PRINCIPLE

The robot's chassis was built entirely from **repurposed cardboard and plastic bottles**. The seed dispenser was engineered from paper — folded into a functional rotating mechanism with no 3D-printed or machined parts. Demonstrated that **functional robotics hardware** can be prototyped with zero-waste materials under competition constraints.

✓ Designed, built & coded full robot independently in 24h

✓ Award-winning recycled-material robot construction

KEY COMPETENCIES DEMONSTRATED

Summary

DISCIPLINE	SCOPE	OUTCOME
Embedded Systems	Arduino C++, 3 sensors, motor control	Fully autonomous navigation & dispensing
Hardware Integration	Breadboard wiring, IR + color sensor fusion	Reliable line-following under competition load
Sustainable Fabrication	Cardboard, plastic bottles, paper dispenser	Zero-waste build — 🏆 Unexpected Expedition Award